

Christian Garry

MSc Scientific Computing | Probability · Statistics · Optimisation | C++/Python
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Education

MSc Scientific Computing & Data Analysis (AI for Engineering) Durham University	Durham, United Kingdom Sep 2025 – Present
• Current average: Distinction (89%). • Focus: probability, statistics, optimisation, numerical methods, Bayesian and regression models, HPC/GPU.	
MEng Electronic Engineering Durham University	Durham, United Kingdom Sep 2020 – Jun 2024
• Upper Second-Class Honours (2:1). • Dissertation: <i>Silicon Carbide JFET CPU</i> — simulation-driven CPU design with instruction-level verification and architectural trade-off analysis.	

Experience

Industrial Tutor Durham University, Department of Engineering	Durham, United Kingdom Sep 2025 – Present
• Led weekly design tutorials, mentoring teams on feasibility-driven decision-making under cost, risk, and performance constraints. • Assessed and provided structured feedback on technical feasibility, assumptions, and justification quality.	
Graduate Communications Engineer Siemens PLC	Hebburn, United Kingdom Sep 2024 – Present
• Implemented and adapted C/C++ communication components (TCP/IP, serial) to support data exchange between simulated systems and PC-based engineering tools. • Built Python-based engineering tooling to structure and query internal technical documentation and codebases using a retrieval-augmented workflow (local LLMs, vector search). • Adapted existing network signal and waveform-generation utilities to support development and integration testing workflows.	

Projects & Research

Silicon Carbide JFET CPU Master's Dissertation	Durham, United Kingdom Oct 2023 – Apr 2024
• Independently designed and simulated a 4-bit CPU using SiC JFET logic in LTspice, targeting computation in extreme temperature and radiation environments. • Constructed a full verification toolchain (C++ compiler, assembler, Python automation) and validated instruction-level behaviour via emulator parity checks and waveform-based state reconstruction. • Investigated architectural trade-offs, race conditions, and storage constraints, explicitly documenting assumptions, limitations, and scalability bottlenecks.	

Key Skills

Probability & Statistics; Numerical Methods; Optimisation; Monte Carlo Simulation; Python; C++;
High-Performance Computing; Linear Algebra

Leadership, Activities & Interests

Research Interests: Time-series modelling, signal validation, regime dependence in financial and physical systems. **Leadership:** Bishops Diocesan College Fencing Team Captain; Durham Fresher Representative (2022–23).
Activities: Durham University Fencing Team; Boxing (Student Fight Night).
Achievements: Weldon le Huray Fencing Scholarship (2020–24); South Africa U17 Fencing Champion; President's Award (Gold).